SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2_AT2 — VISUALIZATION DEVELOPMENT EXERCISE

Course Code:	DSA06	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO2 – Development (BL4)	**Assessment:**	Assessment Tool 2 – Visualization Development
Weightage:	20% of CIA	**Total Marks:**	25 (5 Questions × 5 Marks)
CO2 Statement:	*Analyze and synthesize multiple data views into a coherent, annotated dashboard that communicates a clear narrative insight.(BL4)*		
Student Name:	Jothika M	**Reg. No.:**	192324235
Section / Batch:	SLOT C	**Date:**	30-07-2026

Code of Conduct

I Jothika M(192324235) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Jothika M

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must include at least two coordinated charts sharing a common axis or theme.
- Include at least one annotation calling out a key data point in your dashboard.
- Write a one-sentence narrative takeaway summarizing the insight your dashboard reveals.

Annexure A – VISUALIZATION DEVELOPMENT

Problem 7 — Manufacturing Quality Dashboard (5 Marks)
Data: Defect rate and production volume by line: Line 1 = 2% / 1,000 units, Line 2 = 5% / 1,200, Line 3 = 1% / 800, Line 4 = 8% / 1,500, Line 5 = 3% / 900.
Task: Build a dashboard with a bar chart of production volume and a bar chart of defect rate for the same lines. Annotate the line requiring immediate quality review, and add a takeaway for the plant manager.
Problem 8 — Financial Portfolio Dashboard (5 Marks)
Data: Allocation and annual return by asset class: Stocks = 50% / 12% return, Bonds = 30% / 4%, Real Estate = 15% / 8%, Cash = 5% / 1%.
Task: Build a dashboard combining a pie chart of allocation and a bar chart of returns for the same asset classes. Annotate the class with the best risk-adjusted contribution, and write a narrative recommending a rebalancing action.
Problem 9 — HR Attrition Dashboard (5 Marks)
Data: Headcount and attrition rate by department: Engineering = 120 / 8%, Sales = 80 / 15%, Support =60 / 20%, Marketing = 40 / 10%.
Task: Build a dashboard with a bar chart of headcount and a bar chart of attrition rate for the same departments. Annotate the department needing urgent retention action, and write a one-sentenceexecutive takeaway.
Problem 10 — Environmental Monitoring Dashboard (5 Marks)
Data: Monthly air quality index (AQI) and average temperature: Jan = 80 / 15°C, Feb = 75 / 17, Mar = 90 / 20, Apr = 110 / 24, May = 130 / 28, Jun = 100 / 30.
Task: Build a dashboard with a line chart of AQI and a line chart of temperature on the same month axis(stacked, not dual-axis). Annotate the month where AQI peaked, and write a narrative captionconnecting temperature and air quality trends.
Problem 11 — Sports Team Performance Dashboard (5 Marks)
Data: Wins/losses and points scored: Team A = 18W-6L / 2,200 pts, Team B = 14W-10L / 2,100, Team C =20W-4L / 2,400, Team D = 10W-14L / 1,900.
Task: Build a dashboard with a bar chart of win–loss record and a bar chart of points scored for the sameteams. Annotate the standout team, and write a takeaway comparing win record to points scored.

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT2 — Visualization Development Exercise

CO2 · BT4: Analyze ·

This rubric is used to assess student responses to the CO2-AT2 Visualization Development Exercise, in which students build a small multi-chart dashboard from given data, add at least one annotation, and write a narrative takeaway.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	**25%**	Demonstrates thorough understanding of which chart types to combine, how they relate to each other, and how annotation and narrative should be used to communicate insight.	Good understanding of chart combination and annotation/narrative technique, with only minor gaps.	Basic understanding of dashboard development, but with some misconceptions about which charts to combine or how to annotate them.	Poor understanding of core development concepts; combines charts or annotations that do not suit the data or the story being told.
Explanation	**25%**	The annotation and narrative takeaway are precise, well-justified, and clearly tied to a specific, correctly identified data point or pattern.	The annotation and narrative are clear and reasonably well-justified, with adequate connection to the data.	The annotation or narrative is present but vague, generic, or weakly connected to the actual data shown.	Annotation or narrative is missing, incorrect, or unconnected to the data in the dashboard.
Application	**20%**	Effectively applies multi-chart coordination, annotation, and storytelling techniques to produce a complete, cohesive dashboard for the scenario.	Applies development techniques to produce a workable dashboard, with only minor errors or missed refinements.	Limited application; the dashboard partially addresses the scenario but charts feel disconnected or the story is unclear.	Fails to apply appropriate development techniques; the charts, annotation, and narrative do not work together as a dashboard.
Organization	**15%**	Charts are clearly	Charts are organized and	Basic structure; charts are present	Poorly organized;

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		coordinated (aligned axes or shared theme), laid out logically, with the annotation and narrative placed for maximum clarity.	mostly coordinated, with only minor issues in alignment or layout.	but poorly aligned or the annotation/narrative feels disconnected from the layout.	charts are not coordinated, and the annotation or narrative is missing or misplaced.
Accuracy	**10%**	All chart values, scales, and the annotated data point are completely accurate and correctly reflect the given data.	Mostly accurate, with only minor omissions or a small error in one chart or the annotation.	Partially correct; some plotted values or the annotated point are missing or incorrect.	Largely inaccurate; major errors in plotted values, scales, or the annotated data point.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

CO2-AT-2

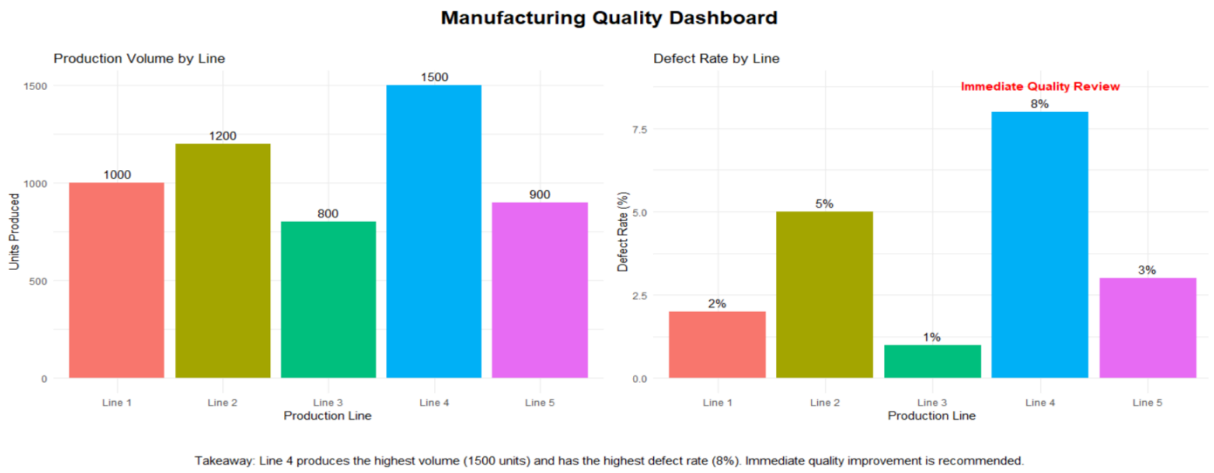
Problem 7 – Manufacturing Quality Dashboard (5 Marks)

Introduction

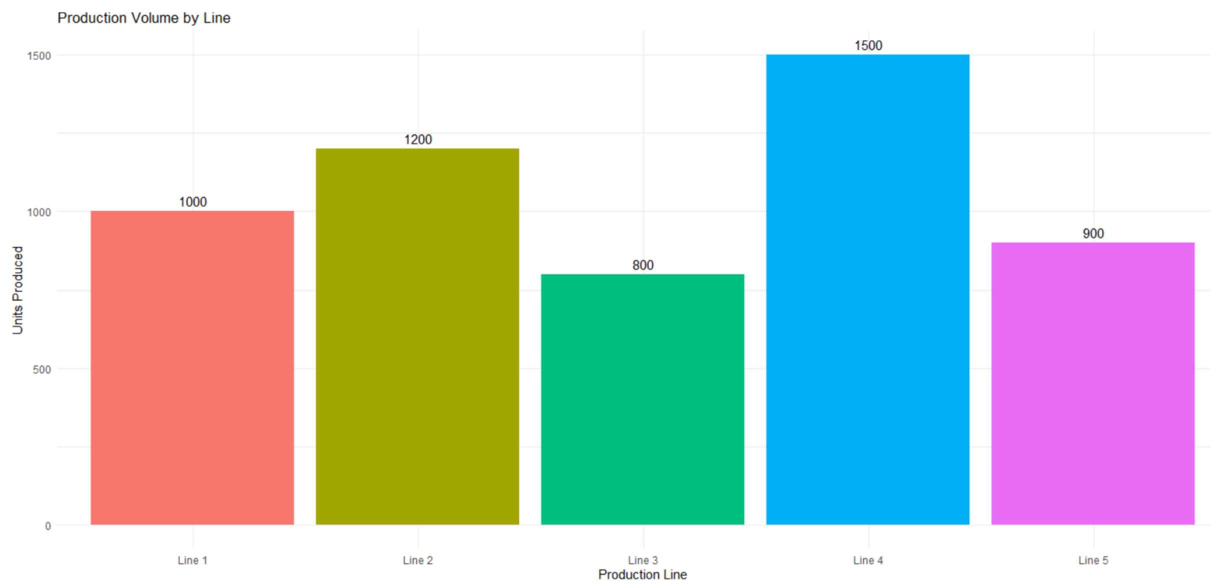
A manufacturing quality dashboard combines multiple visualizations to monitor production efficiency and product quality across different production lines. By displaying both production volume and defect rate together, plant managers can quickly identify which production line requires immediate quality improvement.

The dashboard contains:

Bar Chart 1 – Production Volume by Production Line



Bar Chart 2 – Defect Rate by Production Line



Production Data

Production Line	Production Volume (Units)
Line 1	1,000
Line 2	1,200
Line 3	800
Line 4	1,500
Line 5	900

Quality Data

Production Line	Defect Rate
Line 1	2%
Line 2	5%
Line 3	1%
Line 4	8%
Line 5	3%

Evidence

Line 4 has the highest production volume (1,500 units) while also recording the highest defect rate (8%). In contrast, Line 3 has the lowest defect rate (1%), producing 800 units, indicating better product quality despite lower production output.

Interpretation

The dashboard indicates that Line 4 requires immediate quality review because it produces the largest number of defective products due to its high production volume combined with the highest defect rate. If quality issues are not addressed, the company may experience increased production costs, material wastage, customer complaints, and reduced overall efficiency.

Conclusion

The dashboard enables plant managers to compare production output and product quality simultaneously, helping identify production lines that require immediate corrective actions. It supports better quality control, process improvement, and efficient manufacturing operations.

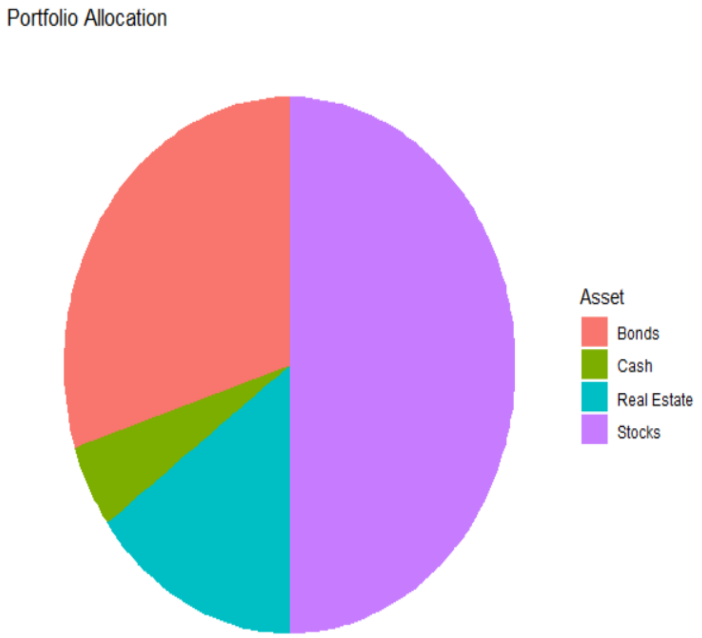
Problem 8 – Financial Portfolio Dashboard (5 Marks)

Introduction

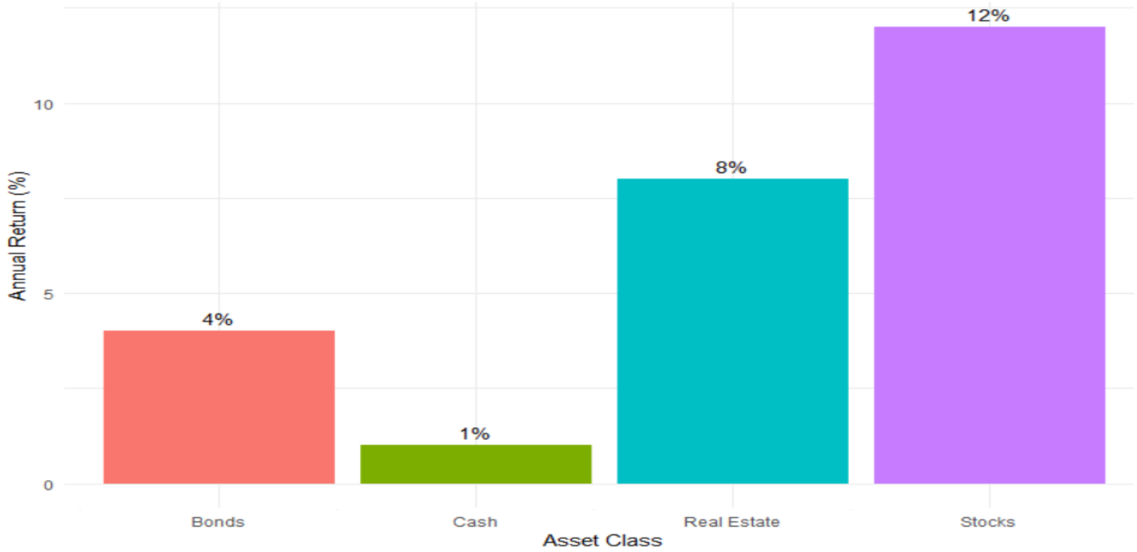
A financial portfolio dashboard combines asset allocation and investment performance into a single view to help investors evaluate portfolio balance and returns. Comparing allocation percentages with annual returns supports better investment decisions and portfolio optimization.

The dashboard contains:

Pie Chart – Portfolio Allocation by Asset Class



Bar Chart – Annual Return by Asset Class



Portfolio Allocation

Asset Class	Allocation
Stocks	50%
Bonds	30%
Real Estate	15%
Cash	5%

Annual Return

Asset Class	Annual Return
Stocks	12%
Bonds	4%
Real Estate	8%
Cash	1%

Evidence

Stocks represent the largest portfolio allocation (50%) and generate the highest annual return (12%). Real Estate provides a solid 8% return with a moderate 15% allocation, while Cash has the lowest allocation (5%) and the lowest annual return (1%).

Interpretation

The dashboard shows that Stocks provide the strongest overall contribution by combining the highest investment allocation with the highest annual return. Real Estate also delivers good returns and can improve portfolio diversification. Cash contributes very little to portfolio growth because of its low return.

Narrative Recommendation (Rebalancing Action)

The portfolio should be slightly rebalanced by reducing a portion of the cash allocation and investing it in Real Estate or Stocks. This adjustment can improve the overall return while maintaining diversification and reducing dependence on a single asset class. Regular portfolio reviews should be conducted to ensure that the asset allocation continues to match investment objectives and market conditions.

Conclusion

The dashboard helps investors compare portfolio allocation and investment performance simultaneously, enabling informed decisions about diversification and portfolio rebalancing. It supports better long-term investment planning by identifying asset classes that contribute most effectively to overall portfolio performance.

Problem 9 – HR Attrition Dashboard (5 Marks)

Introduction

A dashboard combines multiple visualizations into a single view to help managers monitor important business metrics and make quick decisions. In this dashboard, two bar charts are used to compare department headcount and attrition rate.

Headcount

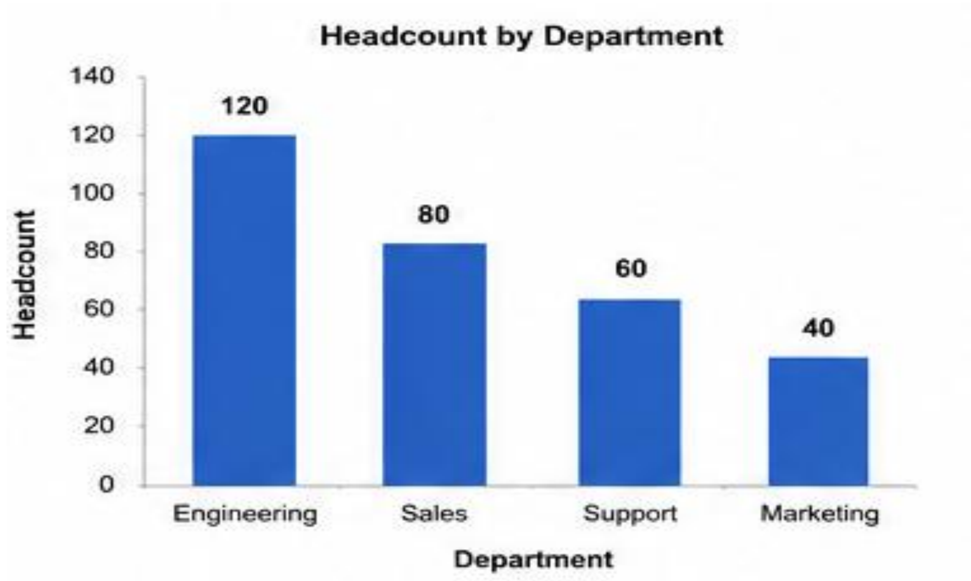
Department	Headcount
Engineering	120
Sales	80
Support	60
Marketing	40

Attrition Rate

Department	Attrition
Engineering	8%
Sales	15%
Support	20%
Marketing	10%

The dashboard contains:

- Bar Chart 1 – Headcount by Department
- Bar Chart 2 – Attrition Rate by Department



- Annotation highlighting the **Support** department
- Executive takeaway below the charts

Evidence

Engineering has the largest workforce (120 employees), whereas Support has the highest attrition rate (20%). Although Support has fewer employees than Engineering, it is losing staff at the fastest rate.

Interpretation

The Support department requires immediate retention strategies because a high attrition rate can increase recruitment costs, reduce productivity, and negatively affect customer service.

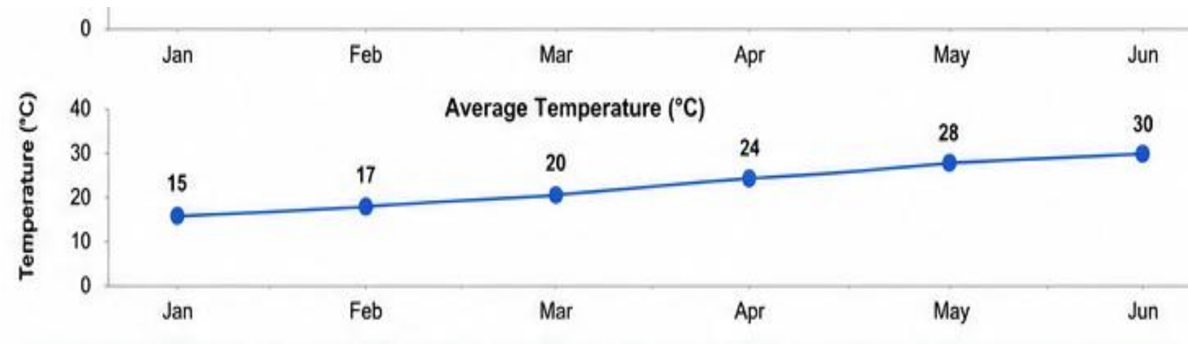
Conclusion

The dashboard enables HR managers to compare workforce size and employee turnover simultaneously, helping identify departments that require immediate attention.

Problem 10 – Environmental Monitoring Dashboard (5 Marks)

Introduction

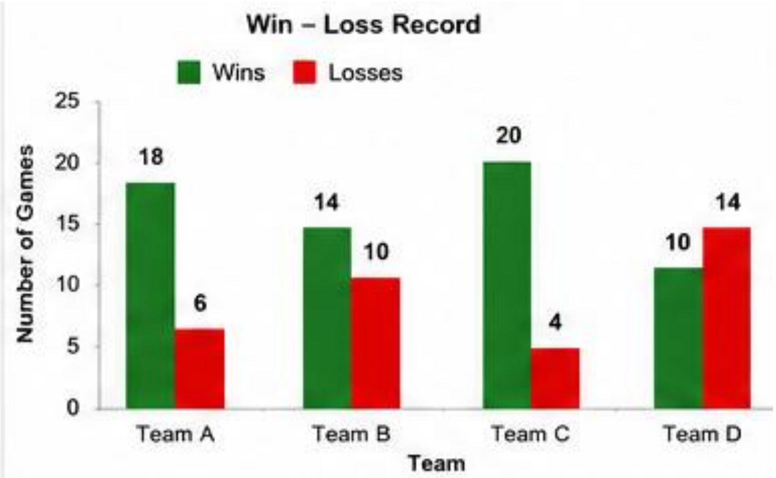
An environmental dashboard combines multiple line charts to monitor changes in environmental conditions over time. Using a common month axis makes it easier to compare trends between different variables.



The dashboard contains:

- Line Chart 1 – Air Quality Index (AQI)
- Line Chart 2 – Average Temperature
- Annotation at the highest AQI value
- Narrative caption explaining the trend

Month	AQI	Temperature (°C)
Jan	80	15
Feb	75	17
Mar	90	20
Apr	110	24
May	130	28
Jun	100	30



Evidence

AQI gradually increases from January to May and reaches its highest value of **130** in May. Temperature also rises steadily from **15°C** to **30°C** throughout the six months.

Interpretation

The dashboard suggests that increasing temperatures are associated with worsening air quality from January to May. Although temperature continues to rise in June, AQI decreases, indicating that other environmental factors may also influence air quality.

Conclusion

The dashboard effectively compares environmental indicators over time and helps identify periods of poor air quality for planning and monitoring purposes.

Problem 11 – Sports Team Performance Dashboard (5 Marks)

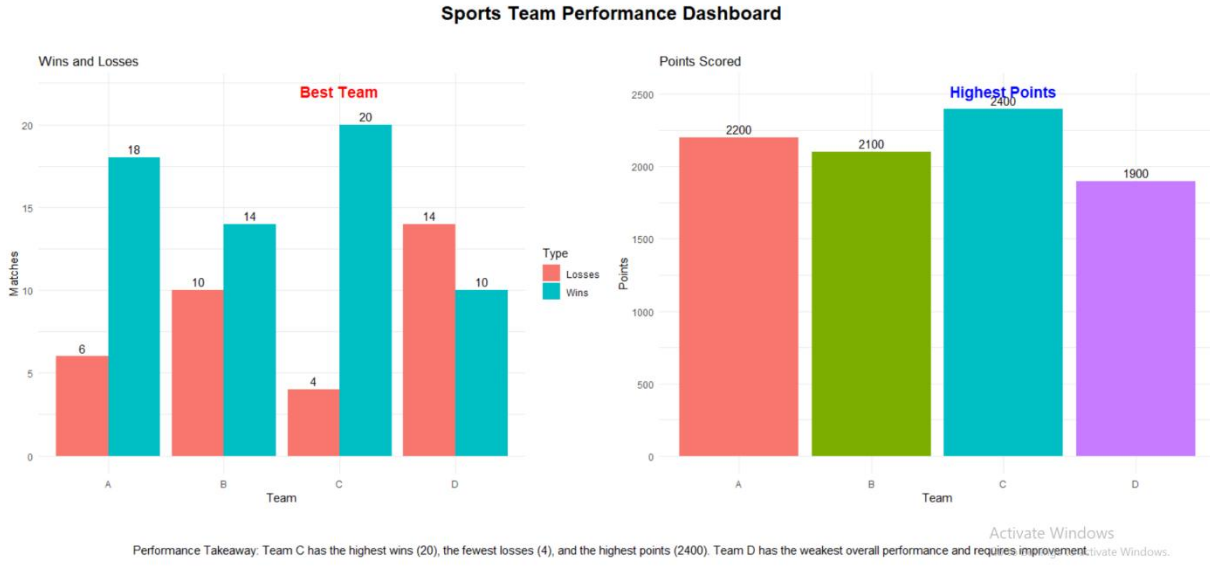
Introduction

A sports performance dashboard combines multiple charts to compare team performance using different metrics. This dashboard evaluates teams based on both their win–loss record and total points scored.

The dashboard contains:

- Bar Chart – Wins and Losses
- Bar Chart – Points Scored
- Annotation identifying the standout team
- Performance takeaway

Team	Wins	Losses	Points
A	18	6	2200
B	14	10	2100
C	20	4	2400
D	10	14	1900



Evidence

Team C has the highest number of wins (**20**), the fewest losses (**4**), and the highest points scored (**2400**).

Interpretation

The dashboard shows that Team C consistently outperforms the other teams across both performance indicators. Team D has both the weakest win–loss record and the fewest points scored, indicating poor overall performance.

Conclusion

The dashboard provides a comprehensive comparison of team performance and supports coaches and managers in evaluating strengths and weaknesses.